

LETS: A Distributed Runtime for Lineage-Bounded Autonomous Systems

System paper and artifact report for version 1.0.10

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Standalone Apache-2.0 implementation at <https://github.com/AstralDeep/LETS>

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Abstract

Autonomous systems increasingly create descendants, delegates, replicas, services, and robots that act concurrently and may remain useful while disconnected. Qualitative authorization alone does not bound the aggregate quantity of effects that a recursively changing population can produce, while a central counter makes disconnected work unavailable. We present Lineage Escrow Transition Systems (LETS), a standalone distributed authorization runtime. A fixed set of stable warden processes owns disjoint shares of a finite, multi-dimensional envelope. Ephemeral subjects receive signed, expiring leases whose resources and capabilities can only be attenuated. Protected effects are state-machine transitions; a warden atomically debits a transition and emits a short-lived, audience-bound receipt that an independent executor verifies and durably claims.

This paper reports the implemented version 1.0.10 runtime rather than the earlier in-memory research kernel. It includes independent SQLite-backed warden processes, signed cluster manifests, authenticated peer HTTP, sequenced exactly-once target credit for rights transfers, branch revocation, canonical signed JSON, externally anchored warden and executor authority, bounded maintenance, an authenticated HTTP API and a strict-decoding, bounded-retry client, authority-safe immutable observation snapshots, quiescence-first terminal restart fencing, pre-cleanup failure-evidence harvesting, operational tooling, and a host-neutral adapter. We give an inductive conservation argument and state the availability cost: a partitioned site may spend only rights it already owns, so safety and local progress are preserved while remote capacity can be stranded. The artifact contains 706 collected unit, integration, security, property, recovery, and end-to-end cases. A final one-host production-profile acceptance used the exact immutable candidate with three hardened wardens; it exercised mTLS and JWT rejection paths, a bidirectional partition, abrupt restart, automatic convergence, an externally anchored executor replay store, stale-restore rejection, and aggregate conservation. The repository also supplies a separately hardened, one-warden-per-host production profile whose admission depends on operator-provided storage, identity, key, PKI, and rollback-domain controls. This is not a claim of Byzantine tolerance, arbitrary live migration, generic exactly-once physical effects, or independent security audit.

Claim discipline. IMPLEMENTED denotes behavior present in the source artifact; MEASURED denotes a reproduced observation with a recorded environment; CHECKED denotes tests, property exploration, or a bounded abstract state-space check; DESIGNED denotes a protocol argument that is not a mechanized proof; and FUTURE denotes work outside version 1.0.10. These labels prevent architecture, tests, and measurements from being reported as interchangeable forms of evidence.

1 Introduction

Agent replication is an authority-allocation problem before it is a copying problem. A child may be created by cloning code, starting a process, delegating a task, launching a container, or instantiating a robot controller. None of those mechanisms determines how much authority the resulting population can exercise. A parent that gives each child its full quota amplifies authority; a shared eventually consistent counter can overspend during a partition; and a central authorization service sacrifices disconnected progress.

LETS treats spawn as one governed transition in a more general lineage lifecycle. Its core object combines: (i) a finite integer resource vector; (ii) qualitative capabilities; (iii) an immutable transition machine; (iv) nested expiries and branch revocation; and (v) distributed escrow held by stable wardens. The system is reasoner-independent. An LLM, symbolic planner, workflow engine, or fixed controller can propose the same transition. Safety depends on complete mediation at a warden and protected executor, not on model compliance.

The component ideas are established. Database escrow partitions rights so local transactions can preserve numeric invariants (O’Neil, 1986); bounded counters extend this idea to eventually consistent systems (Balegas et al., 2015); exactly-once quantity transfer addresses duplicate delivery (Shoker et al., 2015); and finite quota can be recursively delegated (Agudo et al., 2009). Capabilities, attenuated credentials, state machines, and runtime security automata are likewise prior art (Saltzer and Schroeder, 1975; Birgisson et al., 2014; Andersen et al., 2019; Harel, 1987; Schneider, 2000). The contribution is a narrow composition and its operational realization: stable escrow replicas are separated from churn-heavy lineage subjects, protected effects are receipts for funded state transitions, and rights move between wardens through a durable protocol rather than by copying an active subtree.

1.1 Contributions and non-claims

This artifact makes six concrete contributions.

1. IMPLEMENTED A real multi-node runtime in which each warden has an independent process, database, signing identity, monotonic authority anchor, audit archive, and initial escrow share in the production profile. Schema 2 makes peer replay part of the same anchored core authority.
2. IMPLEMENTED A durable lifecycle and transition service that provides root issuance, recursive attenuating spawn, renewal, quiescence, closure, expiry reclamation, branch revocation, and signed executor receipts under atomic local transactions.
3. IMPLEMENTED A signed, sequenced rights-transfer protocol with idempotent target acceptance, bounded reordering, acknowledgements, and bilateral prefix checkpoints.
4. DESIGNED An inductive safety argument for local and global vector conservation, capability attenuation, nested expiry, and receipt mediation, together with an explicit partition-availability trade-off.
5. IMPLEMENTED A bounded observation and restart-evidence lane that publishes one coherent, authority-bound snapshot without request-path database admission, quiesces workload requests before terminal fencing, and permits process replacement only after an exact durable fence proof.
6. CHECKED Reproducible unit, security, property, crash-recovery, bounded-model, package, and real three-node fault evidence, with the evidence types kept distinct.

LETS does not claim that escrow, delegation, state machines, signatures, leases, or hash-chained audit are individually novel. It does not claim Byzantine safety, automatic policy correctness, instant offline revocation, transparent active-subtree migration, or exactly-once physical effects from a receipt alone.

1.2 From research kernel to distributed runtime

The original manuscript described an in-memory Python kernel and a proposed deployment. Version 1.0.10 replaces that implementation claim with a persisted service and preserves the old kernel only as historical material. Table 1 summarizes the boundary.

2 System Model and Requirements

2.1 Entities and trust boundaries

Let W be a configured, bounded set of wardens. Each warden is a stable accounting process with its own durable database and Ed25519 signing identity. Let L be the dynamic set of leases issued to agents or other subjects. Membership in L can grow and shrink recursively; membership in W changes only through an operator-controlled configuration epoch.

An envelope is named by $(tenant, envelope, epoch)$ and fixes an ordered resource schema. For m dimensions, resource values are elements of $\mathcal{R}_m = \{0, \dots, 2^{63} - 1\}^m$ with componentwise order and checked arithmetic. A policy binds that schema to one immutable state machine, lease and receipt time bounds, a clock uncertainty bound, an evidence predicate language, and a transfer gap window. Policies and machines are content addressed.

The trusted computing base consists of each warden process, its policy and configured evidence evaluators, database, clock within the configured uncertainty, signing provider, manifest trust roots, production authority-anchor and audit-archive services, protected executors and their claim anchors, the TLS/identity termination boundary, and the host isolation that prevents subjects from writing those components. Agents, generated code, prompts, memory, discovery documents, request bodies, networks, and other tenants are untrusted. Wardens are trusted but may crash. Messages may be lost, duplicated, delayed, reordered, replayed, or partitioned.

Table 1: Evidence status in version 1.0.10.

Area	Artifact status	Evidence class
Local lifecycle	Durable lease, policy, receipt, audit, idempotency, and outbox transactions	IMPLEMENTED, CHECKED
Multi-node transfer	Signed prepare/accept/finalize/checkpoint protocol over authenticated HTTP	IMPLEMENTED, CHECKED
Partition behavior	Both directed A-B peer links disabled with a durable failed transfer record; dispatcher converged after restoration	MEASURED
Restart behavior	Warden process replaced on its persisted authority unit; anchored state resumed and transfer converged	MEASURED
Conservation	Inductive argument, SQL commit checks, property traces, bounded abstract checker, aggregate e2e assertion	DESIGNED, CHECKED
Rollback fencing	Database-instance and append-only history heads compared through external linearizable anchors	IMPLEMENTED, CHECKED
Observation and restart	Immutable authority-bound snapshots, freshness failure, quiescence-first fence proof, and bounded pre-cleanup evidence	IMPLEMENTED, CHECKED
Production profile	Admission requires one warden per host, mTLS, external providers, independent state/anchor/audit/backup domains, and an immutable image digest; one-host acceptance emulates the controls but not independent failure domains	IMPLEMENTED, CHECKED (one host)
Byzantine wardens	No threshold authorization or replicated-state-machine defense	FUTURE
Active-subtree migration	Free-right transfer followed by target issuance only	FUTURE

2.2 Safety and availability requirements

The principal safety property is that no execution schedule creates resource rights. Spawn and transfer may relocate rights; a transition may move them into consumed accounting; reclamation may return unconsumed rights; no operation may increase the global envelope. Capability and machine authority must attenuate down a lineage, lease expiry must be nested, and an executor must reject an invalid, stale, wrongly addressed, or replayed receipt.

Availability is conditional. A healthy warden should authorize from its local share without a peer round trip. Under a partition, a site can continue only while it has locally escrowed rights, valid leases, a sufficiently certain clock, and a reachable protected executor. Remote rights are not borrowed speculatively. This chooses bounded authority over full utilization during every partition.

3 LETS Design

3.1 Local ownership equation

For warden w , let $\mathbf{S}_w$ be its signed genesis share, $\mathbf{I}_w$ cumulative accepted inbound transfer, $\mathbf{F}_w$ its free pool, $\mathbf{R}_w$ the sum of residual rights in unreclaimed leases, $\mathbf{C}_w$ cumulative consumption, and $\mathbf{O}_w$ cumulative prepared outbound transfer. The persisted local invariant is

$$\mathbf{S}_w + \mathbf{I}_w = \mathbf{F}_w + \mathbf{R}_w + \mathbf{C}_w + \mathbf{O}_w. \quad (1)$$

Every term is a nonnegative vector of the same dimension. The manifest verifies $\sum_w \mathbf{S}_w = \mathbf{B}$, where $\mathbf{B}$ is the global initial budget. Since an inbound credit can exist only for a signed, already prepared source voucher, define the source-owned in-flight vector on a consistent cut as $\mathbf{X} = \sum_w (\mathbf{O}_w - \mathbf{I}_w)$. Summing Eq. 1 yields

$$\mathbf{B} = \sum_w (\mathbf{F}_w + \mathbf{R}_w + \mathbf{C}_w) + \mathbf{X}. \quad (2)$$

Before target acceptance, transferred rights are in $\mathbf{X}$; after acceptance, matching outbound and inbound cumulative terms cancel. This formulation avoids counting a prepared voucher as both source capacity and target capacity.

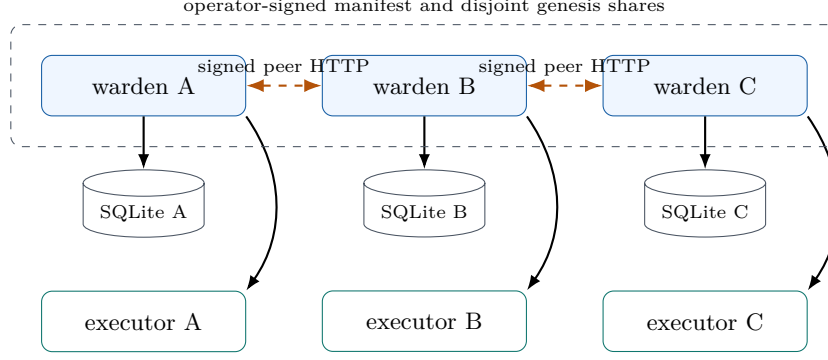


Figure 1: Reference topology. The acceptance deployments instantiate three independent warden processes, stores, and keys. The production profile additionally separates each warden database, monotonic authority anchor, audit archive, and backup domain; its protected executor similarly separates schema-5 replay state from the executor anchor. Ephemeral subjects call wardens but are not escrow replicas.

3.2 Signed leases and recursive attenuation

A lease grant binds tenant, envelope, epoch, lease and lineage identifiers, parent and subject, original allocation, capabilities, policy and machine digests (the latter committing to the initial machine state), ancestor path, branch epoch, issue and expiry times, warden and key identifiers, and a signature. The mutable snapshot contains residual vector, current state, lifecycle status, and monotonically increasing sequence.

Root issuance is a privileged operation: a transport-authenticated identity must hold `lets.admin`, `lets.warden.admin`, or `lets.lease.issue`. It debits the local free pool and cannot request a capability absent from the bound machine; the allocation must contain at least one positive component. Spawn is available to the authenticated parent subject or an identity holding `lets.admin` or `lets.lease.manage`. A child allocation is likewise nonzero. For parent p and child c , the transaction requires

$$\mathbf{a}_c \leq \mathbf{r}_p, \quad K_c \subseteq K_p, \quad (3)$$

$$t_c^{exp} \leq t_p^{exp}, \quad d_c^{policy} = d_p^{policy}, \quad d_c^{machine} = d_p^{machine}. \quad (4)$$

It then atomically replaces $\mathbf{r}_p$ by $\mathbf{r}_p - \mathbf{a}_c$ and creates the child with residual $\mathbf{a}_c$. A request cannot substitute a cheaper or more permissive machine behind a different policy identifier. Expected state and sequence fields provide optimistic concurrency guards. A durable request identifier is unique envelope-wide across mutation types; its full fingerprint makes retries idempotent and rejects a same-key/different-operation or body collision.

Replication has deliberately narrow semantics. It creates a new governed runtime identity and partitions authority. It does not copy bearer tokens, private keys, human identity, process memory, open sockets, uncommitted state, protected data, or signing secrets. Artifact and workspace transfer remain host operations.

3.3 Transition authorization and protected execution

Each protected transition specifies source state, target state, required capability, a componentwise nonnegative resource cost with at least one positive component, and an optional declarative evidence rule. The evidence language is data, not executable policy code; policy and machine digests include its semantics. Evidence can satisfy a guard but cannot mint resources or capabilities.

After the network boundary authenticates the request and the service validates the identity object, authorization performs the following stateful checks and mutations in one local write transaction:

1. tenant/subject/scope binding; lease status, branch epoch, expiry, and clock checks;
2. request-id fingerprint lookup and optional expected state/sequence checks;
3. immutable policy/machine resolution, legal transition, capability, audience, and evidence evaluation;
4. componentwise residual debit and machine state/sequence update;
5. signed receipt, idempotent response, hash-chained audit event, and outbox persistence; and
6. durable commit before returning the receipt.

The protected executor has a separate trust configuration and replay database. It always verifies the exact audience, receipt freshness under bounded clock uncertainty, and signature against a registered warden/key pair that

is currently valid at verification time. Local ExecutorPolicy fields and allowlists can further pin tenant, envelope, epoch, policy digest, machine digest, and warden.

The production-grade SQLite executor library profile uses schema 5. One immutable replay identity binds the exact audience, tenant, envelope, configuration epoch, hash of the complete executor policy, and hash of the complete verification-key registry. Each successful claim appends a digest-chained history record and advances a database-instance-bound claim head and clock floor. A linearizable external compare-and-swap anchor must acknowledge that committed head before the claim returns; an older restore, divergent copy, losing concurrent clone, policy widening, or verification-key substitution therefore fails closed. Live receipt and lease-watermark rows are each pruned in batches of at most 128 (up to 256 rows across the two tables per successful claim), but the authority claim chain is append-only and requires capacity planning for a finite executor epoch. Schema 4 has no promotable external claim history; transition to schema 5 requires a drained, newly initialized authority unit. The repository’s production Compose unit runs wardens, not an executor service; applications must host the executor, its independent state and anchor, and its atomic or idempotent physical-effect integration.

The claim requires the lease sequence to advance and claims both receipt identifier and nonce. This supplies at-most-once authorization consumption. A generic receipt cannot make an arbitrary physical effect exactly once: the executor must bind receipt claim and effect in one domain transaction, make the effect idempotent, or use compensation.

3.4 Lifecycle and revocation

Quiescence disables transition execution without inventing a second policy-machine state. Resume restores actionability if the lease remains valid. Renewal is bounded by the policy maximum and by the parent expiry; a root renewal additionally requires `lets.lease.manage` or `lets.admin`. Version 1.0.10 refuses parent shortening that would strand a live longer-lived descendant. Close and expiry reclamation return only the lease’s own residual, once. Reclamation waits past clock uncertainty and the maximum outstanding receipt lifetime so a receipt cannot remain valid after its authority has been reassigned.

A branch revocation is a signed monotonic epoch for a lineage/branch prefix. Only the branch-owning warden may create it; authenticated peers may relay the owner-signed record. Connected descendants fail closed once the revocation is visible. A disconnected descendant cannot be revoked instantly without communication. Its additional resource exposure is nevertheless bounded componentwise by the residual already present in that branch, and its time exposure is bounded by nested lease expiry plus the configured clock/receipt safety margin under complete mediation. This is an engineering bound, not a claim that a compromised executor or bypass path is contained.

4 Cross-Warden Rights Transfer

4.1 Protocol

For each $(source, target, envelope)$ stream, the source allocates strictly increasing sequence numbers. The protocol is a durable handoff, not a distributed transaction spanning both databases.

Prepare. The source atomically subtracts the amount from its free pool, increments cumulative outbound accounting, assigns the next sequence, and persists a signed voucher. A timeout does not restore the amount. Before target acceptance it remains source-owned in-flight authority in Eq. 2, but it is no longer spendable.

Accept. The target authenticates both the peer HTTP message and the signed voucher, checks tenant/envelope/epoch/source/target/policy binding, and enforces a bounded gap window. Unique sequence and transfer-id bindings prevent alternate payloads. The first valid acceptance credits the free pool and cumulative inbound accounting in one transaction and stores a signed acknowledgement. A business-level duplicate returns the stored logical acknowledgement without a second credit; an exact replay of the signed HTTP envelope is rejected by the durable nonce store.

Finalize and checkpoint. The source verifies the target signature and voucher digest, records finalization idempotently, and advances a contiguous acknowledgement watermark. Once both peers agree on a finalized contiguous prefix, a signed checkpoint permits deletion of individual rows below that prefix. Gaps, conflicting identifiers, backward watermarks, unknown epochs, and untrusted keys fail closed.

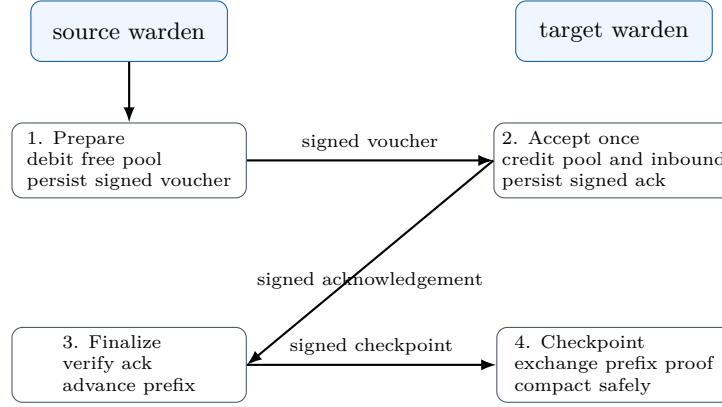


Figure 2: Transfer ownership moves monotonically. A timeout never refunds a prepared source voucher; retries recover the acknowledgement without a second target credit.

Watermark advance and physical history deletion are deliberately incremental: each contiguous- prefix or physical-prune maintenance primitive processes at most 64 transfer-history rows per invocation; direct record and conflict checks are separate. An exact duplicate acknowledgement, voucher, or checkpoint can drive the next bounded watermark or maintenance batch without a second credit or new signature. Branch-revocation materialization and expiry reclamation are likewise limited to 64 leases per transaction; dispatcher and audit-export work use configured bounded batches (64 by default). These limits bound individual WAL/write sets while repeated work converges.

For audit export, each record reaches the idempotent archive before local acknowledgement. The sink-committed prefix is then acknowledged in one reserved authority-serialized transaction per batch, rather than one authority admission per record. A crash before the batch commit leaves the published prefix pending for exact repair from the archive head; no record is acknowledged before its archive write succeeds. Each export cycle also carries a two-second publish budget measured from cycle start: an expiring budget acknowledges the prefix published so far, and follow-up cycles run immediately whenever a cycle was truncated or its snapshot filled the batch, skipping the archive-head call and prefix acknowledgement because the head is exactly the record just acknowledged; every fresh cycle refetches the head, preserving rollback detection. The budget can only take effect between publishes; the archive-head call, one in-flight publish, and the busy-bounded acknowledgement transactions each retain their own five-second allowances, so the fault-free depth-one worst case for a record that just misses a batch snapshot is 36 seconds, and the declared 40-second export staleness bound dominates it while deeper backlogs drain at publish throughput.

IMPLEMENTED A running warden’s bounded peer dispatcher discovers authoritative transfer, revocation, and checkpoint records through durable delivery metadata. It schedules one head per ordered stream, signs HTTP requests, retries failed delivery with bounded exponential delay, finalizes acknowledged transfers, creates checkpoints, and prunes only terminal delivery metadata. Work is not lost if the process crashes between the authority transaction and dispatcher discovery. At-least-once peer delivery remains conditional on connectivity, a live trusted peer, and both signing keys remaining current; idempotent receiving converts redelivery into exactly-once accounting credit, not exactly-once network transmission.

4.2 Failure outcomes

Table 2 states the intended result, including the availability cost.

5 Implementation

The implementation is a typed Python package supporting Python 3.11–3.14 (AstralDeep, 2026). The core depends on PyNaCl 1.6.2 over libsodium for Ed25519; optional server and client extras provide FastAPI/Uvicorn and HTTPX. Development dependencies are resolved by the committed lockfile into a repository-local .venv. Containers build an image-local environment and do not mount the host environment.

Table 2: Transfer and partition outcomes. “Safe” means no duplicated spendable rights under the trusted-warden model; it does not imply liveness.

Failure	Result	Property
Voucher lost after prepare	Amount stays unspendable at source; retry same durable voucher	Safe, unavailable
Acknowledgement lost	Redelivery returns stored acknowledgement; target is not credited twice	Safe, retryable
Out-of-order delivery within window	Target records gap; contiguous watermark advances when missing prefix arrives	Safe, bounded memory
Gap beyond window or conflicting sequence	Request rejected	Fail closed
Target crash after commit	Persisted credit, acknowledgement, and replay state recover on reopen	Safe after recovery
Peer partition	Each warden continues only from local rights; transfer stalls	Safe, partial progress
Signer reaches finite not_after with pending records	Verification continues to fail closed; prepared rights or revocation/checkpoint delivery can remain stranded	Safe, unavailable
Concurrent source database/key clones	Production anchor admits one contiguous history branch and faults the loser; unanchored development can duplicate authority	Fail closed

5.1 Durable storage and atomicity

Each SQLite database owns one envelope. It uses foreign keys, write-ahead logging, synchronous=FULL, a bounded busy timeout, schema/application identifiers, migrations, startup integrity checks, and `BEGIN IMMEDIATE` for serialized writes (SQLite Project, 2026). Checked vector functions are registered with SQLite. Triggers maintain the lease-residual aggregate and validate dimensions; a transaction-level pre-commit check rejects a local conservation violation. Compound foreign keys prevent mixing a policy version with another digest. The authoritative tables cover metadata, warden state, immutable policies, leases, receipts, idempotency responses, revocations, inbound and outbound transfer streams, checkpoints, signed audit, and outbox state. Core schema 2 also owns peer-HTTP nonce claims, their monotonic clock floor, revision, and history digest. An exact schema-1 migration refuses to proceed while any legacy claim remains live. Once all have expired, it binds the legacy artifact digest and clock floor into the new history, advances the new floor to at least the migration time, and never opens the retired file as live state again. The runtime does not make those retired bytes immutable; operators must protect the exact verified backup and migration journal as evidence.

Production opens require an external linearizable authority anchor outside the database rollback domain. A checkpoint binds warden/envelope/epoch, schema, signing-key fingerprint, random database instance, audit head, state revision/digest, and clock floor. Storage reconciles before every transaction and after each committed write; an acknowledged write result is not released ahead of the post-commit compare-and-swap. The database can recover a committed local extension, but an older or divergent restore and a losing clone cannot pass admission against the advanced anchor.

The capacity contract distinguishes logical from physical limits. `max_database_bytes` installs SQLite `max_page_count` on every connection and caps the logical main database; it does not cap the live WAL or shared-memory index. Before an authority write, LETS reserves the emergency filesystem floor plus all remaining main-file growth, a conservative transaction that could write every configured page to WAL, and additional WAL-index growth. It also retains a logical page reserve inside the main-file ceiling. Capacity recovery paths remain available only within this reserve. `SQLITE_FULL`, including when raised by checkpointing, becomes a sticky fail-closed fault; other checkpoint failures are surfaced explicitly. Operators must provide an exclusive quota or dedicated filesystem for the free-space reading to be meaningful. The contract is conservative headroom, not a throughput or retention guarantee.

The storage abstraction admits another backend, but only SQLite has equivalent implementation and fault evidence in this release. The release dependency audit selected PyNaCl/libsodium for the signed path and removed an unused PostgreSQL extra; version 1.0.10 exposes neither a PostgreSQL dependency nor a validated adapter.

The supplied deployment validator and production server admission reject a SQLite runtime that lacks the 2026 WAL-reset safety fix and explicitly reject the withdrawn 3.52 release line. Older SQLite measurements in Section 7 are development-only algorithmic observations, not evidence that such a runtime is production-admissible.

5.2 Canonical signed data and key trust

All content digests and signatures use the documented LETS-CJ/1 profile. Object keys are sorted by Unicode scalar value, arrays retain order, integer values are lossless signed 64-bit values, and output is minimal UTF-8 JSON. Floats, non-finite values, duplicate keys, non-string keys, lone surrogates, and non-canonical base64url spellings are rejected. LETS-CJ/1 is intentionally not RFC 8785 JCS because nanosecond timestamps and resource values may exceed the lossless integer range of common IEEE-754 JSON consumers (Rundgren et al., 2020). Cross-language implementations must use the published conformance vectors and lossless parsing.

Signed-record type tags use the LETS-owned `lets.*` version namespace; the cluster manifest uses `lets.manifest/v1`. Neither the core protocol nor its canonical bytes contain an AstralDeep-specific type namespace.

Ed25519 authority records bind both warden and key identifiers. The trust registry rejects conflicting registrations. Manifest-bound service startup, readiness, and every authority-record signing path require the complete declared clock-uncertainty interval to fit within the signer’s configured not-before/not-after interval; peer receivers independently enforce current-time key validity for protocol records and signed HTTP requests. A new policy registration applies the authority-signing rule before commit. The cluster manifest binds the envelope schema, global budget, exact disjoint shares, peer endpoints, keys, and policies; local bootstrap trust configuration fixes the accepted operator keys and required signature threshold. Manifest verification counts distinct operator key material rather than aliases, requires warden key identifiers and Ed25519 public-key material to be globally unique, and forbids Ed25519 public-key material reuse across operator and online warden roles. Non-loopback plain HTTP endpoints require an explicit development override; production deployment requires TLS in addition to message signatures.

The warden database never stores the private signing seed. Its immutable bootstrap metadata binds the initial warden/key identifier and SHA-256 public-key fingerprint; signing-key identity or public-key material substitution therefore fails startup. Manifest-backed service startup reloads strict JSON and re-verifies the manifest digest, operator threshold, local key/configuration, and complete peer trust before serving.

Version 1 resource limits bound adversarial parser and validation work: at most 256 resource dimensions, 1,024 machine transitions, 64 ancestor-path entries, and evidence expression depth 32 with at most 256 nodes and 256 membership values per node. These are protocol limits, not measured capacity guidance.

5.3 Network planes

The client plane strictly decodes JSON and rejects unknown top-level fields in endpoint request bodies; deliberately opaque evidence and configuration values remain open, and dynamic policy registration performs targeted semantic checks but does not enforce the complete nested OpenAPI Policy schema. The published OpenAPI 3.1 artifact documents closed request schemas for clients, endpoint 2xx response schemas, and application/problem+json errors. Runtime handlers produce those responses, but the bundled client does not validate success bodies against OpenAPI. Identity is derived by an injected authenticator, never accepted from body fields. The bundled static bearer authenticator is development bootstrap tooling and stores only token digests.

Production selects exactly one installed `lets.runtime_providers` entry point. The provider must return the exact tenant/warden, a signer, authenticator, independent authority anchor, independently durable audit sink, and an exact `production_capable` declaration. The built-in raw-seed/static-bearer provider is forbidden. The bundled vendor-neutral provider calls an operator-controlled Ed25519 signer helper without a shell, verifies short-lived EdDSA JWTs, and uses separately mounted file anchor and SQLite audit archive. Those local bindings make the profile runnable; a cloud KMS, PKCS#11 device, OIDC/SPIFFE identity plane, remote linearizable anchor, and managed archive remain operator-installed integrations, not bundled conformance.

Authentication is enforced at the network boundary. Direct calls to service and administrative ports are trusted in-process integration calls, not authenticated remote entry points; an embedding host must keep them inside its TCB.

Peer requests carry method, raw target path/query, body digest, timestamp, nonce, warden, key, and an Ed25519 signature. Schema 2 claims each nonce in an externally anchored core write before handler execution, prunes at most 128 expired claims per request, and advances a digest-bound replay clock floor. That nonce write is not atomic with

the later peer business operation, so a valid request can consume its nonce even if the handler subsequently rejects or fails. Peer signatures authenticate content but do not provide confidentiality or hide metadata. Production serving requires inbound mTLS and HTTPS manifest endpoints; peer-enabled nodes additionally require hostname-verifying outbound TLS with a client certificate. Certificate issuance, revocation policy, and identity lifecycle are deployment responsibilities. TLS and the bundled generic provider's JWT trust are loaded at process startup, so their certificate, key, and trust rotation requires controlled process recreation with a configured overlap interval; the Python TLS path does not perform online OCSP validation. In the bundled profile, mTLS channel admission and JWT or LETS signed-message authorization are independent checks: the JWT is not bound to the presented client certificate, and tenant/scopes are not derived from its subject or SAN. Certificate-bound workload identity requires an operator provider or gateway integration. The generic authenticator requires a syntactically valid jti but does not persist or consume it; the same bearer JWT can therefore be replayed until its expiry plus accepted skew. Revocation and one-time-token semantics likewise require an operator identity or gateway integration.

The operator surface provides liveness, bounded clock-aware readiness, invariant snapshots, audit verification, bounded audit pagination, and metrics for pools, consumption, transfer watermarks/gaps, capacity, and outbox lag. A background exporter reconciles the core audit chain with an independently durable, identity-bound contiguous archive. The exporter waits at most a configured deadline for each sink call; a timed-out call may continue in its single daemon worker, and later publication attempts fail while that worker remains live. Blocked publication, divergence, excess backlog, or stalled progress makes production readiness false. Export acknowledgement and outbox pruning are bounded and resumable. The signed core audit chain itself is immutable and retained; export is rollback/recovery evidence, not permission to delete history.

Production backup creates an exclusively published recovery bundle only after a durable drain and empty peer/audit queues. The bundle contains configuration, the core database, any required legacy replay artifact, the signed manifest, and checkpoint metadata. It intentionally excludes the live authority anchor, audit archive, signer/provider recovery material, identity keys, and PKI; those external dependencies require their own coordinated recovery procedures. Restore begins from a separately trusted live configuration and signed manifest and requires the bundled copies to be byte-identical; it does not adopt lost configuration or perform an epoch migration. It validates a scratch copy, compares it non-mutatingly with the current anchor before publication, moves the old core/WAL/SHM sidecars into quarantine across resumable phases, and removes that quarantine after successful admission. The restored node remains drained until explicit operator activation. These mechanisms fence stale state; they do not make simultaneous copies an endorsed topology. The bundled recovery path covers warden core authority only, not schema-5 executor replay state or its claim anchor. Executor recovery is application-operated: fence effects, snapshot the replay database, retain the anchor separately, and require an anchored reopen.

5.4 Authority-safe observation and terminal replacement

Production health cannot safely bypass authority ordering by reading the SQLite WAL directly: a database commit can become visible before the external anchor compare-and-swap has acknowledged the branch. Version 1.0.10 instead constructs an immutable observation through one priority, authority-safe read admission. The capture binds the database identity, authority lifetime and checkpoint, core revision, audit head and verifier state, runtime readiness, conservation vectors, peer/exporter state, schema hashes, and monotonic capture time into a canonical snapshot digest. Request handling serves that process-local snapshot asynchronously, without another storage admission or shared worker-thread hop. A capture failure immediately makes the retained document unready; age beyond the 15-second bound is returned explicitly as stale rather than hidden behind a transport timeout.

Planned replacement separates preparation from the 30-second process-replacement window. The workload first quiesces at a completed request boundary. The host then validates an exact terminal audit proof and re-inspects the unchanged container before a durable container-clock acknowledgement binds the terminal proof and target identity. Only after that acknowledgement may the host issue SIGKILL; replacement identity, authority lifetime, checkpoint lineage, and completion are then checked within the acknowledged bound. The one-hour release harness retains bounded health, workload, coordination, resource, and failure diagnostics before it removes the scenario volume. These records support offline validation; they do not make the host clock and container monotonic clock numerically interchangeable.

Table 3: Major implementation boundaries.

Module	Responsibility
policy, models	Immutable transition/evidence semantics and strict signed wire records
service	Lifecycle, transition, revocation, transfer, invariant, and audit transactions
storage.sqlite	Versioned schema, durability/capacity admission, repositories, conservation checks, and core replay authority
crypto, canonical	Ed25519 trust and LETS-CJ/1 bytes/digests
auth, api, client, peer	Transport identity, signed HTTP/replay defense, durable peer dispatch, and client/CLI contracts
executor, executor_authority	Independent receipt verification, schema-5 at-most-once claim, and external rollback anchor
authority, audit, recovery	Warden monotonic CAS, independent contiguous audit export, and journaled bundles/restore
observation	Priority coherent capture, incremental audit verification, immutable freshness-bound publication, and terminal audit proof
runtime, providers	Selected signer/identity/anchor/audit provider contract and bundled generic profile
manifest	Operator-signed cluster bootstrap and exact genesis-share validation
integrations	Host-neutral lifecycle port and optional AstralDeep scope profile

5.5 Module decomposition

6 Safety Argument

This section is a protocol-level inductive argument, not a machine-checked refinement proof. The bounded checker and executable tests discussed in Section 7 provide separate, finite evidence.

6.1 Local conservation

Lemma 1 (local preservation). If Eq. 1 holds before a committed operation, each implemented operation preserves it componentwise.

Argument. Root issuance moves a vector from F_w to R_w . Spawn moves a vector between parent and child terms inside R_w . An authorized transition moves its cost from R_w to C_w . Close or reclamation moves a residual from R_w to F_w . Quiescence, resume, renewal, and revocation change status or time metadata but not resource terms. Prepare subtracts from F_w and adds to O_w . Target acceptance adds the same voucher amount to both F_w and I_w . Finalization and checkpointing change no vector. Retried idempotent operations return the prior result; rejected or rolled-back operations change no authoritative state. Checked arithmetic prevents wraparound. $\square$

Theorem 1 (global conservation under the failure model). If the signed manifest exactly partitions B , all wardens begin in a locally valid state, peer keys are trusted, and every committed operation is one of the operations above, then Eq. 2 holds on every consistent cut despite crash, retry, duplication, reordering, loss, or partition.

Argument. Lemma 1 preserves every local equality. A target increments I only after verifying a matching signed prepared voucher; uniqueness constraints prevent a second credit. Therefore cumulative inbound never creates an unmatched source debit. Summation and the manifest share equation yield Eq. 2. Message failures affect whether a prepared term moves out of X , not whether it is duplicated. $\square$

The theorem excludes a compromised warden signing nonexistent vouchers, manifest trust-root compromise, and direct database modification. Production admission adds a linearizable external anchor: two database/key clones may race, but only one contiguous history branch can advance and the loser faults before returning an acknowledged authorization. This fencing argument depends on anchor availability, independence from the database snapshot, and correct compare-and-swap semantics; it is not Byzantine replication.

6.2 Lineage and transition properties

Lemma 2 (attenuation). Along every issued lineage edge $p \rightarrow c$, the child’s capability set is a subset of the parent’s, the immutable policy/machine digests are equal, the child expiry is no later than the parent’s, and the child allocation is removed from the parent in the same transaction.

Argument. These are explicit preconditions and writes of the spawn transaction. Subject binding prevents an unscoped sibling from invoking spawn on another lease. The policy digest and machine digest are signed into the grant. $\square$

Lemma 3 (transition soundness). Under complete mediation and uncompromised trusted code, an executor accepts a receipt only for a committed transition whose source state, capability, evidence, vector cost, policy/machine digests, audience, sequence, and validity interval passed the warden checks.

Argument. The debit, state/sequence update, signed receipt, idempotency response, and audit record commit atomically. The executor independently verifies all signed bindings and atomically claims the receipt. In production its external claim anchor must acknowledge the append-only head before the claim is returned. A missing, invalid, stale, divergent, or replayed receipt/claim state fails closed. $\square$

Lemma 3 says nothing about the semantic adequacy of a configured cost or evidence predicate, nor about effects reachable through a bypassable executor.

7 Evaluation and Reproducible Evidence

The evaluation asks whether the implementation satisfies its protocol obligations under bounded and adversarial schedules. It is not a claim of external security certification or broad workload performance.

7.1 Evidence layers

Table 4: Verification layers and what they can establish.

Layer	Reproduced status	Scope and limitation
Static checks	Ruff check and 132-file format check, strict mypy on 34 source files, uv lock check, container build-context policy, and release-verifier checks passed	Ruff formatting/lint and strict mypy check source consistency; neither proves protocol safety.
Test suite	704 passed; 2 Docker e2e deselected	Unit, integration, security, property, concurrency, crash, and recovery behavior; finite executions only.
Compatibility/package	required Ubuntu Python 3.11–3.14 and Windows Python 3.11/3.14 jobs passed; two byte-identical wheel, sdist, and deployment-bundle builds; Twine 7.0.0, check-wheel-contents 0.6.3, exact archive parity, and isolated wheel/sdist import and CLI checks passed	One additional interpreter and isolated artifact/import checks; not a platform certification.
TLA+ TLC	passed; 6,776 distinct states	Scalar three-warden escrow/transfer/receipt kernel; finite and not a source-code refinement.
Bounded checker	passed; 101,245 states	scalar budget 3; three wardens; at most three leases, two transfers, and two receipts; unit actions; depth 9; abstracts signatures, clocks, SQL, and HSM labels.
Production acceptance	passed in 36.900 seconds	Exact-digest candidate under one-host Linux Compose; mTLS, hardening, partition, restart, convergence, and anchored executor checks, not independent failure domains.
Sustained production profile	passed for 3,600 seconds; 362 health samples	One seeded exact-candidate run with bounded partitions and replacements; not a capacity benchmark or independent-failure-domain trial.
Microbenchmarks	100 operations/path; FULL durability	Development measurements only; no cross-system baseline unless reported with environment and repetitions.

7.2 Bounded formal exploration

The artifact supplies a finite TLA+ specification of scalar escrow, transfer, receipt, and replay semantics (Lamport, 1994). TLC completed its configured TypeOK and conservation checks with 40,057 generated states, 6,776 distinct states, maximum depth 14, and an empty queue. A separate dependency-free executable abstract model explores root issuance, recursive spawn, consumption, close/reclamation, sequenced transfer preparation, reordering, acceptance, finalization, receipt claim, and duplicate delivery. Under scalar budget 3; three wardens; at most three leases, two transfers, and two receipts; unit actions; depth 9, it explored 101,245 states, 318,558 transitions, and 66,072 idempotent self-loops; zero invariant violations. A fault-injection mode that credits an already accepted transfer found a three-action prepare, accept, duplicate-accept counterexample at depth 2. The latter is a checker-sensitivity test, not evidence that the production implementation contains that fault. Both models abstract signatures, clocks, SQL, HTTP, and key labels; neither is an unbounded proof or a refinement check against the runtime. Executable service trace conformance is tested, not proved as a refinement mapping.

Property tests independently generate checked-vector inputs and randomized lifecycle traces. Security cases target privilege boundaries, policy substitution, request-id collision, evidence type confusion, concurrent parent overdraw, transfer identity/sequence conflicts, signature spelling, peer key validity, executor races, and canonical wire parsing. Recovery cases open a database after rollback, committed transaction, and abrupt process exit.

The release candidate’s serial non-e2e gate ran with ResourceWarning promoted to an error: 704 passed; 2 Docker e2e deselected. The two Docker e2e cases are reported separately so an unavailable Docker daemon cannot be hidden inside the core-suite result. Required compatibility CI produced required Ubuntu Python 3.11–3.14 and Windows Python 3.11/3.14 jobs passed. Two clean, archive-derived package builds produced byte-identical artifacts; Twine, wheel-content and RECORD validation, exact source/archive parity, isolated base/server/client wheel and sdists imports, CLI startup on Python 3.14, Bandit 1.9.4 at medium/high severity, and pip-audit 2.10.1 passed. These are bounded tool results, not an external security certification. Hosted CI for acceptance source commit 0d3f5879723ce74b597b846fe909daa1d1086609 completed green. The wheel was 235,760 bytes with digest sha256:292e98eb33d1203f8da6242c8cecdd38a0dbccdef6efeab7c60332eb58fc77d4; the source distribution was 218,558 bytes with digest sha256:5c44407f92be55d13ff9ef58cbf54be6fbbaa557e0486b95b3db94025bc43109; and the exact 16-file deployment bundle was 80,019 bytes with digest sha256:3325b8bf70a3a1ecd8d7bbb882f90d2a52027ff26a5fffb72cc48865162c77915.

7.3 Frozen 1.0.0 storage measurements

MEASURED The reviewed baseline used Windows 10 build 19045; Python 3.14.0; SQLite 3.50.4; AMD64; 32 logical CPUs and the release PyNaCl/libsodium backend. Each main path ran 10 warm-up and 100 measured operations with perf_counter_ns and nearest-rank percentiles. Every writing path kept SQLite WAL synchronous=FULL and one atomic transaction per stage; verify-only has no write. The clean committed benchmark worktree recorded git_worktree_dirty=false at 9f6ba18315c1a1ba854373053f7bd92e18343e9b; its metadata cross-references the clean acceptance source-tree digest. That commit differs from the accepted source commit only in retained evidence files. The single-host in-process measurements exclude HTTP, network, container scheduling, external authority/audit services, and independent failure domains. Its SQLite 3.50.4 runtime is not production-admissible under the release gate. The reviewed machine-readable baseline is retained under benchmarks/baselines.

Table 5: Frozen 1.0.0 development-path latency and wall-time rate. The 100-sample p95 values are indicative, not capacity guarantees.

Path	Median (ms)	p95 (ms)	Rate (op/s)
Warden authorize	8.2877	9.3887	119.24
Transfer prepare	7.48085	8.6475	130.57
Transfer accept	6.9838	7.8025	137.74
Transfer finalize	6.484	7.615	151.56
Transfer round trip	21.08745	23.1221	46.48
Executor verify only	0.17805	0.1829	5,577.09
Executor verify and durable claim	4.41765	5.8287	219.34
Four-worker authorize contention	56.4619	69.5167	49.02

Authorization includes validation, evidence/policy evaluation, Ed25519 signing, audit, invariant maintenance, and

durable commit. The transfer round trip is three FULL transactions across two stores and excludes HTTP/network time. The four-worker row uses independent leases in one single-writer database; latency includes lock wait, and its 69.5167-ms p95 shows why the result must not be generalized to arbitrary write concurrency.

IMPLEMENTED The final optimization pass removed an unconditional full `PRAGMA foreign_key_check` from every successful write while retaining per-connection foreign key enforcement, deferred-constraint rejection at commit, and full startup/explicit diagnostics. MEASURED At 10, 100, 500, and 1,000 receipt rows, the frozen full-durability authorization medians were 6.96635, 7.17425, 8.48515, and 10.52645 ms. The historical pre-hardening observations are retained for engineering context but are not a controlled comparison; authority-anchor, recovery, replay, capacity, and audit hardening all changed before the frozen measurements.

IMPLEMENTED Frequent invariant snapshots now read a trigger-maintained lease-residual aggregate while startup and explicit diagnostics retain full reconciliation. At 10, 100, and 1,000 lease rows, the frozen aggregate medians were 1.3121, 1.29235, and 1.2494 ms. Their near-flat values over this bounded range are not an asymptotic or capacity guarantee; the historical legacy row-scan observations predate the reusable harness and are not treated as a controlled comparison.

A read-only keeper-connection experiment was rejected: four order-alternated trials produced a 1.139237 experimental-to-baseline median ratio, a 13.92% penalty. The release therefore retains per-operation connections. The final profile was dominated by the durable SQLite connection/commit lifecycle; no further source change was justified by this bounded measurement without weakening transaction or durability semantics. The fresh exact dependency graph contains 48 packages in a 315,852-byte lockfile. The runtime cryptographic path is PyNaCl/libodium with CFFI support; the lock contains no cryptography package.

7.4 Hardened production-profile acceptance

The final runner recorded passed in 36.900 seconds against the exact immutable candidate digest `sha256:a2bc64cb516f22c3cc71e7c0ed0b48f618ee125344b28b5ea3be9859a8fe622c` (local content ID `sha256:bb84503ad40196d2dc1301ef12cf3774c919a4d8c9954cae57be0a9ff836cd99`). The 4,490-byte machine-readable record has digest `sha256:ef5a47026e23ebc4c5149021b3db7365fce2975d98d5a2ea3dcbedb2520da00b`. The runner recorded a clean 175-file snapshot with digest `sha256:686a5382ffe645d8665af3f40789167fc7afdec106cba9e2f6c7248ff9a25ae3` at acceptance source commit `0d3f5879723ce74b597b846fe909daa1d1086609(dirty=false)`. All three wardens ran SQLite 3.53.2 and agreed on signed manifest digest `sha256:15d78e1690e38b5ddda94e8a2d6f7da9331557dfce120be0fc374d1a3b3f5337`.

The transport checks required mTLS and rejected a missing client certificate, an untrusted client certificate, a missing JWT, and an expired JWT. The generic-production provider's external signer was invoked 21, 9, and 4 times across wardens A, B, and C. Each warden ran nonroot with a read-only root filesystem, read-only configuration/trust/PKI/signer mounts, hardened temporary storage, bounded resources and logs, distinct state/anchor/audit domains, and no live backup mount. All three core audit chains verified; their independent archives contained 6, 5, and 1 records.

The scenario disabled both directed A-B peer links after preparing transfer sequence 1, leaving a durable failed delivery record. It replaced warden A's process, restored the links, and observed the production dispatcher converge with two delivered records, no pending records, and no prepared transfers. Cumulative inbound and outbound transfer were both 10 units, and the final aggregate conservation check passed. A separate executor profile used independent state and anchor domains, advanced anchored claim sequence 1, rejected a duplicate receipt after reopen, and rejected a stale database restore.

This scenario establishes that the hardened path exercised the exact candidate rather than an in-process simulation. It ran three wardens in one Linux Compose project, so it does not establish independent physical failure or rollback domains, geographic independence, behavior under arbitrary schedules, long-duration stability, Byzantine tolerance, or external certification.

7.5 Sustained production-profile soak

MEASURED The same clean source and immutable candidate completed its sole unweakened 3,600 seconds measurement window in 3,657.799 seconds of harness wall time. The run completed 559 workload cycles, all 362 scheduled health samples, 27 bounded partition episodes, and 3 quiescence-first planned replacements. It recorded zero health-cadence deadline misses and zero HTTP request retries. Independent workload, authority, conservation, resource, checkpoint-lineage, and terminal-fence admission all passed after the embedded release

verifier reconstructed the raw evidence.

The retained 24,066,673-byte record has raw digest `sha256:5977a16f48127e641c43ed2d585598aad25595e97534c1506a4ed5bd2e29d245` and canonical payload digest `sha256:d67162f8dd721ae6a7a1e66abc884d7a47615ad89c8d70fbe52764e6dcebcf30`. This is one deterministic, seeded, one-host Compose run. It exercises sustained fault scheduling and bounded observability but does not prove a failure rate, throughput capacity, independent physical failure domains, WAN behavior, arbitrary schedules, or Byzantine tolerance. The signed-tag release workflow runs a separate mandatory soak before promotion rather than relabeling this local evidence.

The first signed-tag attempt, v1.0.5, correctly stopped during that separate soak when one warden’s oldest pending audit record reached 15.132881691 seconds against the declared 15-second bound. The exporter was still progressing, but per-record local acknowledgement amplified authority-provider admissions under sustained traffic. Version 1.0.6 batches only the local acknowledgement transaction while preserving archive-first ordering and exact crash repair. The failed v1.0.5 run is diagnostic evidence and is not counted as a successful release soak.

The second signed-tag attempt, v1.0.6, completed its full unweakened soak workload – 264 cycles, 362 of 362 required health samples, 26 partition episodes, and three authenticated fenced replacements with zero deadline misses – and then failed closed in independent evidence admission on exactly one raw observation captured during an injected partition. The runtime legitimately reports a bounded class-only volatile dispatch error that outlives its durable failed row once that row is delivered or superseded, but the host and embedded release verifiers required a simultaneous durable retry row for every volatile error. Version 1.0.7 binds the sanitized durable retry summary to the count of undelivered failed records instead of to the volatile error, keeping every exact-schema, bounded-token, and readiness-equality rejection. Replaying the retained v1.0.6 evidence through the corrected verifiers admits all 1,080 raw observation snapshots; the failed run itself remains diagnostic evidence only.

Because this was the third consecutive release stopped by a verifier predicate stricter than a state the runtime legitimately produces, version 1.0.7 also closes three adjacent admission-parity defects found by an adversarial audit of every observation predicate against the producing runtime code: the pre-first-cycle dispatcher startup state after a planned restart, a transient exporter error before the first acknowledged batch resets the volatile success marker, and an audit-error recovery record whose six-decimal timestamp could not equal the three-decimal sample it must match exactly. Each fix is bound by deterministic acceptance and mutation tests against both the host validator and the extracted release-workflow verifier.

The third signed-tag attempt, v1.0.7, confirmed those admission fixes held and failed on a known residual of the 1.0.6 batching change: under partition-driven bursts, batch acknowledgement let a record queued just after a cycle’s batch snapshot wait out that cycle plus the entire next one, and warden B’s oldest pending record reached 15.274 seconds against the then-declared 15-second export bound while the exporter was fault-free and progressing – the same signature as the v1.0.5 failure. Version 1.0.8 responds twice. A two-second publish budget with immediate, overhead-free continuation cycles under truncation or full batches keeps typical cycles short and backlog drain bounded by publish throughput. And because worst-case analysis of the remaining allowances (archive-head call, one in-flight publish, and busy-bounded acknowledgement transactions, each lawfully up to five seconds, plus the poll interval) yields a 36-second fault-free depth-one supremum for a record that just misses a batch snapshot, the export staleness contract itself was re-derived: the runtime now declares a 40-second bound dominating that supremum, the host and release-workflow verifiers pin that exact value, and the in-container validator enforces whatever bound the runtime declares – decoupled from the unchanged 15-second health-cadence limit that the old value had been inherited from. Sustained per-operation latencies near the five-second deadlines remain storage degradation that fails closed by design. The failed v1.0.7 run remains diagnostic evidence only.

The fourth signed-tag attempt, v1.0.8, completed its full unweakened soak with every prior fix holding – no admission, stall, or evidence defect across 169 cycles and all health samples – and failed only host adequacy: the throughput floor required the workload to average at most 15 seconds of active time per cycle, and the runner paced 15.37. Retained evidence across the four release soaks shows fault-free hosted-runner pace ranging from 10.1 to 19 seconds per cycle, so that constant also sat inside lawful environmental variance with zero margin. Version 1.0.9 recalibrates the not-stalled floor to 25 seconds of active time per cycle, dominating the observed range while the separate semantic cycle floor continues to guarantee absolute workload coverage; replaying the retained v1.0.8 evidence against the recalibrated floor passes every adequacy check. The failed v1.0.8 run remains diagnostic evidence only.

The fifth signed-tag attempt, v1.0.9, failed in yet another orchestration-side check while the workload was healthy: a resource sample’s in-container identity probe, which requires exactly one serving process, landed inside a helper subprocess’s fork-to-exec window and observed two identical server command lines. Version 1.0.10 filters candidates whose parent is itself a candidate – a genuine duplicate server descends from the init shim and is never filtered – and

confirms any remaining anomaly across bounded rescans before failing closed, with deterministic synthetic-procfs tests covering both the transient and the genuine-duplicate directions. The failed v1.0.9 run remains diagnostic evidence only.

8 Interoperability Profiles

LETS is an authorization/accounting substrate, not an agent discovery protocol, invocation transport, artifact registry, identity provider, or telemetry backend. Integration therefore uses explicit profiles and keeps discovery separate from authority.

Table 6: Standards and draft composition. “Profile” means a mapping boundary, not bundled conformance.

Concern	Existing standards and drafts	LETS relationship
Discovery and description	A2A Agent Cards (A2A Project, 2026), OASF (AGNTCY Contributors, 2026), Agent Spec (Amini et al., 2025), ARD v0.9 draft (ARD Project, 2026)	Map immutable IDs, advertised skills, endpoints, and artifact references into host policy inputs. Advertisement never grants a lease.
Invocation	MCP 2026-07-28 (Model Context Protocol Maintainers, 2026), A2A tasks/messages (A2A Project, 2026)	Carry a host operation to the adapter. A protected effect still requires a LETS receipt at the executor.
Workload identity	SPIFFE 1.15.2 (SPIFFE Project, 2026), OIDC or mTLS	Inject verified transport identity and key trust. The release includes an authenticator interface, not a bundled SPIFFE control plane.
Artifact and build provenance	OCI (Open Container Initiative, 2025), SLSA 1.2 (SLSA Community, 2025), in-toto 1.2 (in-toto Project, 2026)	Host stores immutable artifact descriptors and attestations; LETS binds their identifiers or evidence digests where policy requires.
Composition and inventory	SPDX 3.0.1 (SPDX Workgroup, 2024), CycloneDX 1.7 (OWASP Foundation, 2025)	Supply machine-readable SBOM/AI-BOM evidence; LETS does not invent another BOM.
Provenance and events	W3C PROV-O (Lebo et al., 2013), CloudEvents 1.0.2 (CloudEvents Authors, 2022)	Export audit/outbox observations for analysis. Exported events are not authority and cannot repair a warden pool.

The host-neutral ReplicaAuthorizer accepts lifecycle IDs, resource vectors, capabilities, TTLs, evidence, and immutable policy references. It deliberately excludes credentials, process memory, owner identity, sockets, and opaque agent state. A profile capability set is an upper bound; unknown mappings fail closed.

AstralDeep is one optional consumer, not a dependency of the core. The included adapter maps only the six declared AstralDeep tool scopes to explicitly configured LETS capabilities and transitions. It imports no AstralDeep module and writes no AstralDeep table. AstralDeep retains its Keycloak/RFC 8693 identity path, owner isolation, permissions, PHI and egress checks, confirmations, and audit. This same separation supports non-agent domains such as robotics, logistics, infrastructure, healthcare workflows, manufacturing, or economic limits: only resource dimensions, machines, evidence rules, and adapter mappings change.

9 Security Analysis

9.1 Attacks addressed

The runtime rejects unscoped self-issuance from the free pool, cross-tenant operations, child capability amplification, policy/machine substitution, residual overdraw, stale expected sequences, duplicate-key JSON, floating-point signed data, alternate base64url spellings, conflicting transfer IDs or sequences, target/source substitution, expired keys, excessive clock uncertainty, wrong receipt audience, and durable nonce/receipt replay. Warden, executor, and peer

replay stores persist nondecreasing observed-time floors and reject wall-clock rollback beyond the configured tolerance or declared uncertainty, including after reopen. A large forward clock jump can instead expire authority early and reduce availability. Concurrency tests exercise one-winner behavior for parent debit, command idempotency, and executor claim. In production, warden and executor anchors additionally reject stale restores, divergent histories, database-instance substitution, and the losing side of a concurrent-clone compare-and-swap race. The executor replay identity rejects a changed policy or verification-key registry even when its public IDs are reused.

Hash-chained signed audit makes row modification or deletion detectable when the complete chain and trusted key are available. The production warden checkpoint binds the current audit head, and the exporter requires an independently stored, contiguous, identity-bound archive head before readiness. An older core database behind the warden anchor fails storage admission. Core/archive divergence instead faults the exporter and makes readiness false; mutation ingress must therefore honor readiness-aware routing because individual handlers do not all gate writes on exporter status. The bundled archive is still operator-controlled infrastructure: it is not third-party witnessing, public transparency, trusted timestamping, or protection against an attacker who rewinds or compromises the database, anchor, and archive together.

9.2 Residual risks and explicit boundaries

Table 7: Version 1.0.10 boundaries.

Boundary	Consequence and required control
Trusted wardens	A compromised warden can sign invalid authority from its assigned share. Use host isolation and protect keys; threshold/Byzantine designs are future work.
One envelope per SQLite DB	Scale by sharding independent envelopes across processes. A PostgreSQL backend requires its own conformance and fault evidence.
Anchor failure domain	Production safety requires a linearizable, durable anchor outside the database snapshot. Co-rollback or compromise of both defeats stale-state fencing.
Restore clones	The anchor admits one contiguous branch, but concurrent copies can race and destroy availability. Fence the original before restore.
Clock dependence	Expiry and key validity rely on bounded uncertainty. Excessive uncertainty fails closed and can reduce availability.
Clock rollback	Durable floors plus production anchors block stale/forked database rollback, not a dishonest clock declaration or rollback of the anchor too. Large forward jumps can expire authority early.
No live epoch rollover	Drain into a new envelope/epoch; automated in-place membership and key rotation are not present. A finite online-key expiry is a hard drain deadline.
No active-subtree migration	Transfer free rights, then issue at the target. Do not copy a live lease as authority.
No generic effect transaction	Receipt claim is at-most-once. Bind it to an idempotent or transactional domain effect.
Finite executor epoch	Schema-5 claim history is append-only. Provision dedicated capacity and drain into a fresh database/anchor before exhaustion.
Capacity boundary	The SQLite cap is logical main-file size; safe physical admission depends on an exclusive enforced filesystem/quota and conservative WAL/SHM reserve.
External services	KMS/HSM, PKI revocation, identity control plane, remote anchors, and managed audit retention are operator integrations, not bundled service guarantees.
Evidence semantics	A valid signed sensor report can still be false or semantically insufficient. Domain trust anchors and policy review remain necessary.
Privacy	Lease paths and audit records reveal organizational activity. Minimize fields and protect/export them under domain retention policy.
Plaintext state	SQLite contains sensitive authority, lineage, and audit metadata. POSIX mode 0600 is best effort; Windows/container ACLs, volume encryption, and backup protection are operator controls.
Embedding boundary	Direct in-process service/admin calls are trusted ports. Expose only the authenticated API or reproduce its identity/scope checks.

Finite manifest key intervals are availability boundaries. Version 1 evaluates key validity at verification time and does not require lease or receipt expiry, or the open-ended retry horizon of peer records, to fit before `not_after`. Once the declared clock interval reaches that boundary, local readiness and every authority-record signing path fail closed; verifiers also reject otherwise-unexpired receipts and undelivered transfer, revocation, or checkpoint records. The bundled audit verifier also applies current-time key validity, so full-chain verification becomes unavailable after the online key’s `not_after`; raw archived signatures need separate archival verification tooling and retained-key policy. Active version 1 online warden keys should therefore normally omit finite `not_after`. If one is configured, treat it as a hard drain deadline: before it, stop new authority work, close or reclaim live leases, wait out outstanding receipts, prove all durable transfer and peer-delivery state terminal, and fence the old epoch before starting a separately initialized replacement epoch. Automated in-place key rotation is not implemented. The included

raw seed-file signer is portable development bootstrap machinery and is rejected by production admission. The generic production provider verifies an external helper’s response against the manifest key, but helper immutability, private-key custody, and any KMS/HSM assurance remain deployment controls.

10 Related Work and Novelty Boundary

Agent Contracts combines multidimensional budgets, lifecycle semantics, recursive delegation, and conservation (Ye and Tan, 2026); Agent libOS provides identity, parent-child lineage, capabilities, checkpoints, and audit for long-running agents (Zhang, 2026). These works invalidate broad novelty claims for hierarchical agent contracts or secure agent runtimes. LETS instead targets the distributed operational semantics of a population-wide hard effect vector during partitions, with stable wardens rather than every descendant serving as a counter replica.

Provenact studies policy-state serializability for concurrent effects (Peng and Wu, 2026). Fixed Ceiling studies authorization continuity and attenuating authority across agent evolution (Zhang and Zhang, 2026). Faramesh places a protocol-neutral authorization boundary before execution (Fatmi, 2026), while AgentBound issues verifiable governance receipts bound to delegation and behavioral artifacts (Kaul et al., 2026). LETS shares their emphasis on execution-time mediation and verifiable decisions; its narrower distinction is conserved, partitioned quantity across concurrently acting lineage branches and cross-warden handoff.

Empirical work such as Agent Matrix motivates treating replication as a concrete systems risk rather than a purely hypothetical capability (Zhang et al., 2025). LETS does not try to detect whether a model “intends” to replicate. It makes spawn an ordinary authorized operation that cannot create net resource authority. This turns a model-behavior question into a mediation and accounting obligation, conditional on complete coverage of protected effects.

The stable-warden design inherits the utilization trade-offs of escrow and bounded counters (O’Neil, 1986; Balegas et al., 2015). Rights placed at an isolated site cannot be consumed at another site until connectivity returns. Dynamic rebalancing, predictive placement, and broader WAN evaluation are optimization questions layered above the invariant, not reasons to relax it.

11 Operational Guidance

A production deployment starts from one operator-reviewed manifest that exactly partitions every budget dimension and binds all initial keys, HTTPS endpoints, policies, and epoch metadata. Run one warden per independently provisioned failure domain. Its state database, authority anchor, audit archive, backup store, read-only trust/configuration, PKI material, and signing/identity provider are distinct named parts of one authority unit, with state, anchor, audit, and backup on separate protected mount roots. The live warden cannot mount the backup domain. The root filesystem is read-only, the process is unprivileged and capability-free, and request concurrency, accept backlog, resources, logging, temporary storage, and shutdown are bounded. These container controls are implemented deployment assumptions, not proof that an operator’s storage controller, ACLs, quota, certificate lifecycle, or failure domains satisfy them.

The supplied production Compose/storage profile is admitted only on Linux/amd64 and Linux/arm64; Docker Desktop on Windows or macOS is suitable for smoke testing, not as a claimed durability or failure boundary. Its bundled SQLite/file-anchor/SQLite-archive provider requires dedicated local Linux filesystems with POSIX locking, WAL shared-memory locking, atomic rename/link, and durable file and directory fsync. NFS, SMB/CIFS, desktop-shared, object, overlay, and RAM-backed mounts are outside that profile, and one state directory must never be attached to two hosts.

Operators should monitor readiness, DB/WAL/SHM and archive capacity, write/anchor/signer latency, busy errors, clock uncertainty, lease and receipt expiry, transfer gaps and watermarks, peer and audit-export lag, executor claim capacity, invariant status, and audit continuity. A source voucher is never manually refunded after timeout. Backups are drain-and-stop recovery bundles; restores are journaled, checked against the live anchor/archive, and remain drained until explicit activation. Idempotency, executor claim history, and signed core audit history are not deleted by hand.

Production startup performs full SQLite integrity and foreign-key scans plus anchor admission before listener startup. Audit reconciliation begins at startup and gates readiness rather than listener creation. The synchronous cold-start work therefore scales with retained state; the orchestrator start window must be measured against the deployment’s worst case, or an undersized window can cause a restart loop.

The repository includes a production validator and an mTLS profile acceptance harness. Validation can check exact paths, ownership/mode constraints, immutable image references, provider loading, signer-helper placement, Compose hardening, and stated capacity arithmetic. It cannot establish the truth of a storage controller’s fencing, a filesystem’s durability semantics, or organizational key/identity procedures. Release automation builds or safely resumes a uniquely addressed multi-architecture candidate, resolves its immutable digest, runs scans and production acceptance against that candidate, and only then promotes release tags and artifacts. This is a provenance-bound project release gate with reproducible Python/deployment artifacts, not a claim that the OCI image was independently rebuilt and byte-compared, third-party certification, a warranty of every deployment, or permission to deploy a mutable tag.

The single-writer SQLite design is appropriate for an independently owned envelope with bounded write concurrency, not an assertion that one process should serve arbitrary fleet scale. Scale-out uses more independent envelopes today. A different store must reproduce transaction isolation, vector checks, uniqueness, recovery, replay, and transfer tests before it can replace SQLite in the trusted path.

12 Limitations and Future Work

The principal internal-validity threat is the gap between protocol arguments and executable refinement. The artifact includes a finite TLA+ kernel and a separately executed abstract checker, but there is no machine-checked refinement connecting every SQL transaction, anchor CAS, and crash point to Eq. 2. The bounded checker covers only one small finite state space. Independent third-party security/audit review and multi-language canonicalization interoperability are still required.

The reported Docker runs are composite scenarios on one host. They verify independent operating system processes and storage volumes, including the production-profile control paths where reported, but not independent failure domains, heterogeneous architectures, real WAN clocks, sustained partitions, rolling upgrades, or long-duration load. Future evaluation should add repeated seeded fault campaigns, protocol-version skew, disk-full and I/O failure, clock excursions, TLS and external identity profiles, and multi-host latency and throughput distributions.

Engineering priorities are threshold authorization for compromised-warden tolerance, safe configuration-epoch rollover, public or third-party audit witnessing beyond the implemented operator-controlled anchor/archive, a validated PostgreSQL adapter, hardware-backed provider evidence, policy analysis, and workload-informed rights placement. Active subtree migration requires a separate handoff protocol that proves a single owner for every live lease and outstanding receipt; it must not be approximated by copy-and-delete.

13 Conclusion

LETS version 1.0.10 is a real distributed, multi-node runtime for governing recursively changing populations under finite effect envelopes. It separates stable escrow wardens from ephemeral subjects, partitions rights rather than duplicating them, authorizes funded state transitions atomically, and requires independently verifiable receipts at protected executors. Cross-warden transfer tolerates retry, duplicate delivery, bounded reordering, restart, and partition without double credit under the trusted-warden model. In the production profile, linearizable warden and executor anchors fence stale or forked local authority, while bounded maintenance, explicit capacity reserves, mTLS/provider admission, contiguous audit export, and journaled recovery turn those safety assumptions into operator-visible fail-closed boundaries.

The central result is deliberately conditional and operational: a site can continue during a partition only from authority already escrowed there. Capacity elsewhere may be stranded. That is the cost of preserving a hard population-wide bound without consensus on every action. The artifact turns the earlier abstract proposal into testable software while retaining honest boundaries around Byzantine faults, anchor co-failure, deployment-specific services, migration, external effects, policy correctness, and the absence of independent third-party certification.

A Operation Delta Table

Spawn has zero aggregate ΔR because it subtracts the child allocation from the parent and adds the same vector to the child in one transaction. A duplicate accepted transfer has all-zero deltas because the stored acknowledgement is returned.

Table 8: Componentwise deltas for Eq. 1. A zero denotes no vector change.

Operation	ΔF	ΔR	ΔC	ΔO	ΔI
Issue root x	$-x$	$+x$	0	0	0
Spawn child x	0	0	0	0	0
Execute cost c	0	$-c$	$+c$	0	0
Close/reclaim r	$+r$	$-r$	0	0	0
Prepare transfer x	$-x$	0	0	$+x$	0
Accept transfer x	$+x$	0	0	0	$+x$
Finalize/checkpoint	0	0	0	0	0

B Artifact Reproduction

All Python commands use the repository-local environment:

```
uv sync --all-extras --frozen
uv run --frozen ruff check .
uv run --frozen ruff format --check .
uv run --frozen mypy src
uv run --frozen pytest -m "not e2e" '
    -W error::ResourceWarning --cov=lets --cov-report=term-missing
$image_name = "127.0.0.1:25109/astraldeep/lets"
$image_digest = "sha256:" + '
    "a2bc64cb516f22c3cc71e7c0ed0b48f618ee125344b28b5ea3be9859a8fe622c"
$env:LETS_PRODUCTION_ACCEPTANCE_IMAGE = "$image_name@$image_digest"
uv run --frozen python deploy/production/run_acceptance.py
```

The bounded checker is reproduced with:

```
uv run --frozen python -m formal.model_checker
```

The retained mutation record uses `--inject-duplicate-credit-fault`; its expected nonzero exit and minimal counterexample show that duplicate credit violates conservation. The retained TLC record includes the tool version, JAR digest, exact command, configuration/specification digests, queue exhaustion, and state counts. On Windows, the one-command runner is:

```
powershell -NoProfile -File formal/run_tlc.ps1
```

It downloads the pinned TLA+ v1.8.0 JAR into an ignored repository-local directory, verifies its SHA-256 and byte count, and uses the preinstalled Java runtime; it installs no system software.

The frozen 1.0.0 benchmark suite uses the repository-local Python environment:

```
uv run --frozen python -m benchmarks.run
uv run --frozen python -m benchmarks.profile_scaling
uv run --frozen python -m benchmarks.profile_invariants
uv run --frozen python -m benchmarks.profile_anchor
```

Reviewed host-specific JSON and CSV are retained below benchmarks/baselines; generated profiles and scratch results remain ignored.

The paper build uses a digest-pinned TeX Live container and writes auxiliary files below `paper/build`:

```
make paper
make paper-check
make paper-render
```

The production acceptance runner requires an exact name-at-digest candidate, materializes ephemeral PKI and provider inputs, validates its controlled local targets, starts the hardened three-warden topology, waits for health checks and directed fault proxies, exercises mTLS, partition, restart, convergence, and executor-anchor cases, writes machine-readable evidence, and stops containers in a finally path. It is a one-host Linux acceptance, not evidence of independent physical failure domains.

C Receipt Verification Checklist

A protected executor should reject unless all of the following hold:

1. the wire record is strict and canonical, with a supported type and no unknown fields;
2. the registered warden/key pair is currently valid at verification time under clock uncertainty;
3. the Ed25519 signature covers the complete receipt payload;
4. every tenant, envelope, configuration-epoch, policy-digest, machine-digest, and warden pin configured by local executor policy matches;
5. the complete executor policy and authenticated verification-key registry match the immutable schema-5 replay identity;
6. the executor audience is exact and the receipt is fresh;
7. the lease sequence strictly advances the durable executor watermark; and
8. the receipt identifier and nonce are claimed once, the append-only claim head advances, and an independent external anchor acknowledges the exact committed head.

Only after this checklist should the executor enter its application-specific effect transaction. The application must still make that effect idempotent or bind claim/effect atomically in its own domain.

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